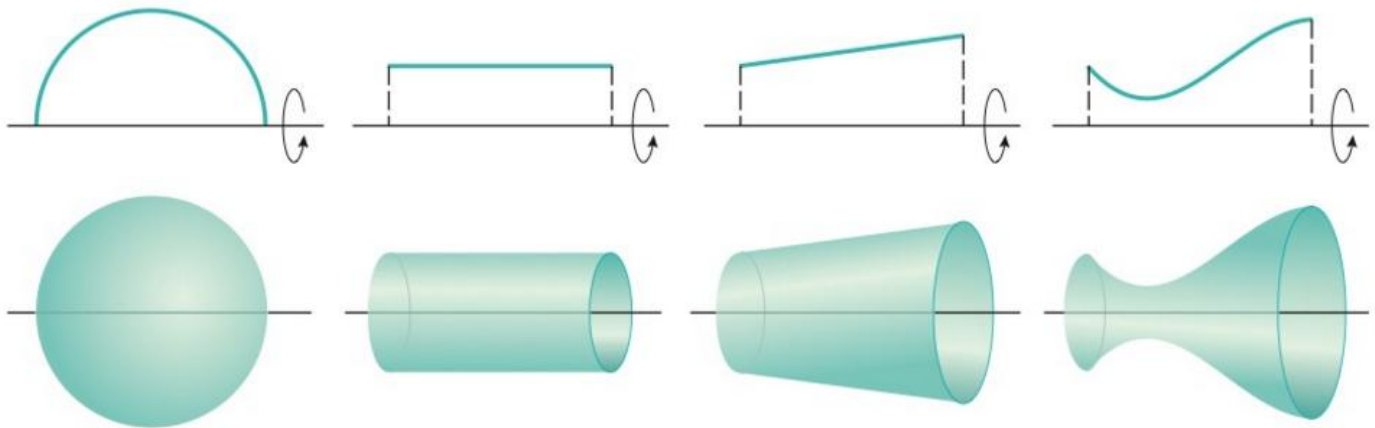


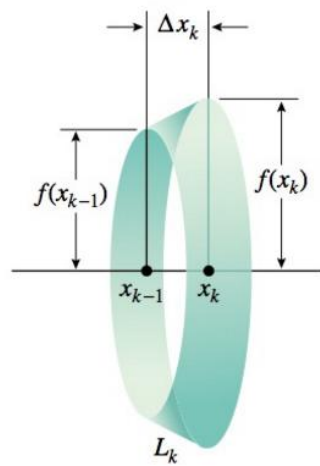
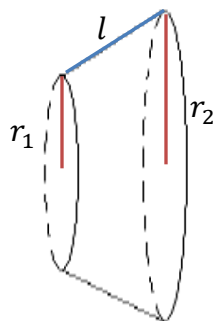
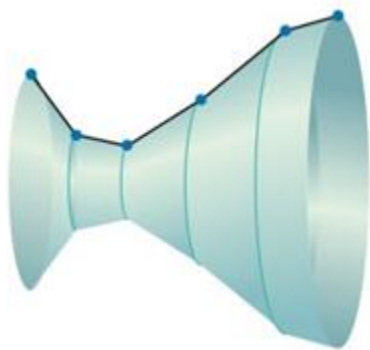
1.5 Surface Area for Solids of Revolution

General Idea:

When a curve is revolved to form a solid we can try to determine the resulting solid's lateral surface area (surface area excluding the bases of the figure).

**The Specifics:**

Suppose we wish to revolve a curve C around an axis and find the surface area of the resulting solid.



Definition: Suppose that $f(x)$ is nonnegative and has a continuous derivative on $[a, b]$, we define the surface area of the solid obtained by rotating the curve $y = f(x)$ where $a \leq x \leq b$ about the **x -axis** as:

We note that:

$$S = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx = 2\pi \int_a^b y \sqrt{1 + [f'(x)]^2} dx = 2\pi \int y ds$$

where ds is the arc length differential (see section 2.4.1 for definition of arc length differential).

Thus, if the curve is described as $x = g(y)$ where $c \leq y \leq d$ then the formula is:

Definition: Suppose that $x = g(y)$ is nonnegative and has a continuous derivative on $[c, d]$, we define the surface area of the solid obtained by rotating the curve $x = g(y)$ where $c \leq y \leq d$ about the **y -axis** as:

Thus, if the curve is described as $y = f(x)$ where $a \leq x \leq b$ then the formula is:

Note: In these definitions we are not able to put limits of integration on the integrals involving the arc length differential because we do not know limits on the s .

Example 1: Find the exact surface area of the figure obtained by rotating the curve $y = x^3$ where $1 \leq x \leq 2$ about the x -axis.

Example 2: Set up, *but do not evaluate*, an integral which yields the exact surface area of the figure obtained by rotating the curve $y = x^3$ where $1 \leq x \leq 2$ about the y -axis.

Example 3: Set up, *but do not evaluate*, an integral which yields the exact surface area of the figure obtained by rotating the curve $y = \frac{1}{4}x^2 - \frac{1}{2}\ln(x)$ where $1 \leq x \leq 2$ about the y -axis.

Example 4: Determine the exact area of the surface generated by revolving $y = \sqrt{25-x}$ where $9 \leq x \leq 16$ about the x -axis.